

TASK-11

8-8-2026

Topic: **AWS VPC subnet configuration**
(Public and Private server)

Step 1: Create a VPC:

1. Open AWS Console.
2. Go to VPC service.
3. Click Create VPC.
4. Name: my-vpc.
5. **IPv4 CIDR: 10.0.0.0/16.**
6. Click Create VPC.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

vishi vpc

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	vishi vpc	Remove tag

Add tag

You can add 49 more tags.

Cancel

Preview code

Create VPC

✓ You successfully created vpc-05562d74fcf13568f / vishi vpc

Step 2: Create a Two Subnets:

1. Go to VPC → Subnets.
2. Enter Name: public-subnet and private-subnets.
3. Enter IPv4 CIDR block: 10.0.1.0/24.
4. Click Create subnet.

Create subnet Info

VPC

VPC ID
Create subnets in this VPC.

vpc-05562d74fcf13568f (vishi vpc) ▼

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference ▼

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.1.0/24 256 IPs

< > ^ v

Tags - optional

Key	Value - optional	
Name	private subnet2	Remove

✓ You have successfully created 2 subnets: subnet-0555c35f9eb2f341a, subnet-078b164f44b7cfc70

Subnets (2) Info

Find subnets by attribute or tag

Subnet ID : subnet-0555c35f9eb2f341a Subnet ID : subnet-078b164f44b7cfc70 Clear filters

Last updated less than a minute ago Actions Create subnet

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	private subnet2	subnet-078b164f44b7cfc70	Available	vpc-05562d74fcf13568f vishi ...	Off	10.0.1.0/24
☐	public subnet1	subnet-0555c35f9eb2f341a	Available	vpc-05562d74fcf13568f vishi ...	Off	10.0.0.0/24

Step 3: Create an Internet Gateway:

1. VPC → Internet Gateways.
2. Click Create internet gateway.
3. Name: my-igw
4. Click Create.
5. Actions → Attach to VPC.
6. Choose my-vpc.
7. Click Attach.

Internet gateways (5) [Info](#)

Find internet gateways by attribute or tag

Actions

Create Internet gateway

< 1 >

The following internet gateway was created: igw-08e1e306a7834a3a1 - vishi vpc igm. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

gw-08e1e306a7834a3a1 / vishi vpc igm

Actions

☐	vishi vpc igm	igw-08e1e306a7834a3a1	Attached	vpc-05562d74fcf13568f vishi vpc	964376925119
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Step 4: Create a Route Table:

1. VPC Route Tables
2. Click → Route table
3. Name : Public -rt
4. Select my-vpc
5. Click create

Create route table [info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the Internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

public rt

VPC
The VPC to use for this route table.

vpc-05562d74cf13568f (vishi vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q public rt X Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create route table](#)

✔ Route table rtb-08445c850a1b62881 | public rt was created successfully. X

rtb-08445c850a1b62881 / public rt [Actions](#)

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0 X	Internet Gateway	-	No	CreateRoute
	Q igw-08e1e306a7834a3a1 X			Remove

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

Step 5: Associate Route Table with Subnet:

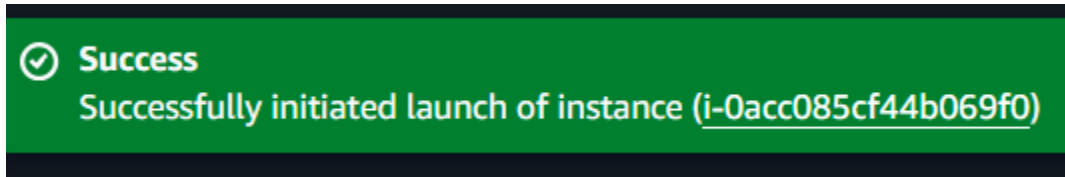
1. In public-rt.
2. Go to Subnet Associations.
3. Click Edit subnet associations.
4. Select public-subnet.
5. Click Save

Routes	Subnet associations	Edge associations	Route propagation	Tags	Related resources
Explicit subnet associations (2) Edit subnet associations					
Find subnet association					
Name	Subnet ID	IPV4 CIDR	IPV6 CIDR		
-	subnet-0caefefc11574a0e8	172.31.48.0/20	-		
-	subnet-0414d6e5e989edac3	172.31.80.0/20	-		

Create the new instance in EC2:

- 1) Public
- 2) Private

☐	public ec2	i-059db0050c94babe4	Running	t3.micro	3/3 checks passed	us-east-1b	-	44.212.10.13
☐	vishi	i-06e1f42d3ef16da70	Stopped	t3.micro	-	us-east-1d	-	-
☐	private ec2	i-0acc085cf44b069f0	Running	t3.micro	3/3 checks passed	us-east-1b	-	-



In the public ec2 it showing the public ip address

Instance summary for i-059db0050c94babe4 (public ec2) Info

Updated less than a minute ago

Instance ID
i-059db0050c94babe4

IPv6 address
-

Public IPv4 address
44.212.10.13 | [open address](#)

Instance state
Running

Private IPv4 addresses
10.0.0.188

Public DNS
-

In the private ec2 it not showing public ip address

Instance summary for i-0acc085cf44b069f0 (private ec2) Info

Updated less than a minute ago

Instance ID
i-0acc085cf44b069f0

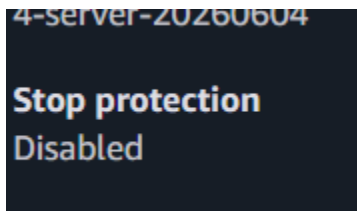
IPv6 address
-

Public IPv4 address
-

Instance state
Running

Private IPv4 addresses
10.0.1.155

Public DNS
-



Open the terminal

```
PS C:\Users\Karthik Reddy> cd downloads
PS C:\Users\Karthik Reddy\downloads> ssh -i "kittu.pem" ubuntu@44.212.10.13
The authenticity of host '44.212.10.13 (44.212.10.13)' can't be established.
ED25519 key fingerprint is SHA256:4Hhdi2N1zdmnNY311W6ZtiJNRCzhdc05Ra1/D1hbz08.
This key is not known by any other names.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.212.10.13' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul  8 04:43:46 UTC 2026

System load:  0.01           Temperature:   -273.1 C
Usage of /:   30.6% of 6.61GB Processes:      119
```

Create the file:

Command : **touch file name**
vi file name

```
ubuntu@ip-10-0-0-188:~$ touch vishitha.pem
ubuntu@ip-10-0-0-188:~$ vi vishitha.pem
```

Open new tab in the terminal

Command for see the key: **cat file name.pem**

```
Install the latest PowerShell for new features and improvements! h

PS C:\Users\Karthik Reddy> cd downloads
PS C:\Users\Karthik Reddy\downloads> cat kittu.pem
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAq/D/oHPmKelNVVreTsBognJdF5dW0VhwcPHTkyRKsTeaXx3f
yErtDLrWk9ISS/J+JRtki4KF37w900E6nYxv/67JFyb26SofHCd52gtNiVeRMnb3
kAA2yinRQd79SQt+qgCEKGptQBhU2IY+7LdMmK5hyMRfEhxbkwSnIKNRGbflNQOD
jpwCjSj+QfrsGiURrNdAY1a00ZKpEtQ8sek5WxX14a92TNNvciDhZMbfcJPFuU5e
36dwixQBUFraDZ/tavarfI2FJ+qfPW/HL0lNW8YA04qsgBMUX1FZjyNCZtT8/KPI
Xz4sLkcAh4iwsDE2iPMkLU3i1RgN402rpdYzlwIDAQABAoIBAHWELaMcNA8nR1T0
c4dN20B3vArYw/UT7L4z2tDx+B8iCt/B8mbaVyThOrHz3c+M4Q0JqPH1dvzRkuKi
hkaSayZNV2H8nrwErxi2HoGaU9y6PiJIz7Hbg5cvd7fW6jUWST12dt2MyDkLqEoJ
j06Jhc/BOFJZbUVannhgsL4NAexZ9oI3G5iTJJj+rbPMLZhHJLxrAocZ7bpQwUpS
FyZEqAJbbly4FTmETzpJxuU8Tf2Jtt+7SZoWzNUS/vSZFmiXJ0h245Jg8gxouiaY
23PrJDPildGMF7Nm0aFE1aWVlVGi3BibgkCwaKNzzyHzllTNNFVc0fuA9N8glMhH
0xWJ1pkCgYEA0o2PezK0rXDtgCTohIXRgvqYlXxiKHx1W4QiRV0+T5Ki93ApA6/f
ORbUKtCVY1jLZUYEGr5+njbmqa8kWoP+AT2ztYt/FB7yZiPKuWeTMFfomRldCN+N
1g4OYM4qXHeDCmklpEnJRRWwhRRzRvQKH6j3wMzoqreZjyXC6HObMCgYEA0Q3k
8VtsKuN8HDRAQs2SMqT0c1mZ0t7MftVPS+4ziHENLY78A0gCyJuFZqhiRh/+Iqwk
zpPcXUhdZTYZSDvSpqGHubyFAB+6jwxeSxgF/OyZfbgZb0GLyviKz91xp52DE6cm
RVhTgFf455Hs0wmj3VCsjpu+VkvRe2DxCH7J5o0CgYEAAnNesoRYldumfUYys1EMm
zY3E6d3s9Lzh15k30jp6NyZBVVbshn2VJ960j2wvTAElaGpe111+0naNHquDxhV6
ugvveXX/ZMqkwKc61R+tiH6B+sHlBr4Y1vJDjaMl9RllPSYgKgiUN8WT2mLweFN
D00+3tpwtWTZJlP7lHoc3cGyAUoP1df7r9xHLjyZgpv2VU0Ag7NbU0JPYi8e0g
R4IYTaojCb/+Suazav1P8v+3kcKfI00Civ6syc5FhYAbK6Q0w1PEVUD+kSL0qLDJ
UjaNdb/bswX3f4102xW/i0eC1b0wfEb/gsbCnGu4ViL7dDESTRqoW9/9XVu8BAXk
47YCpQKBgALLPkh1eDTTKy6qhSoG6EaQL+pR3lvu4916LehbgHvaUYftQnA9zCFV
2Gb9XlhQoJshucOL8Rql+3F4Qwa98LNYvRquZaHpw8KezYnaZKpjHeE00ZwE2wrn
RReKdczKZaLRsNg0BpHlmFq0V6H0MnUEhmfp09dps9ShGVrLqIi8
-----END RSA PRIVATE KEY-----
PS C:\Users\Karthik Reddy\downloads> |
```

Copy the private key and paste it the back tab

!bm :

After that enter

```
ubuntu@ip-10-0-0-188:~$ vi vishitha.pem
ubuntu@ip-10-0-0-188:~$ chmod 400 "vishitha.pem"
ubuntu@ip-10-0-0-188:~$ ssh -i "vishitha.pem" ubuntu@10.0.1.155ssh -i "vishitha.pem" ubuntu@10.0.1.155C
ubuntu@ip-10-0-0-188:~$ ssh -i "vishitha.pem" ubuntu@10.0.1.155
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

```
Last login: Wed Jul  8 05:07:47 2026 from 10.0.0.188
ubuntu@ip-10-0-1-155:~$
```


To exit from there

Command : **exit**

```
ubuntu@ip-10-0-1-155:~$ exit
logout
Connection to 10.0.1.155 closed.
```

To check the update :

Command : **sudo apt update**

```
ubuntu@ip-10-0-1-155:~$ sudo apt update
0% [Connecting to us-east-1.ec2.archive.ubuntu.com (2600:1f18:5c55:4c00:7447:9a97:fa56:cd92)] [Connecting to securit^
C
ubuntu@ip-10-0-1-155:~$ ping google.com
PING google.com (142.251.16.139) 56(84) bytes of data.
```

It was not updating it is in the private server

```
ubuntu@ip-10-0-1-155:~$ exit
logout
Connection to 10.0.1.155 closed.
ubuntu@ip-10-0-0-188:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
57 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-0-188:~$ |
```